



FORCE AND PRESSURE -01

Class 08 - Science

1. A push or pull on an object is called [1]
 - a) Push-pull
 - b) Pressure
 - c) Force
 - d) Gravitation
2. Which of the following is not true regarding friction. [1]
 - a) Friction helps a ship move through water
 - b) Friction helps us walk
 - c) Friction produces heat
 - d) Friction wears down machine parts
3. Pressure is- [1]
 - i. Scalar Quantity
 - ii. Vector Quantity
 - iii. Both Scalar and Vector Quantity
 - iv. Tensor Quantity
 - a) option (iii)
 - b) option (ii)
 - c) option (i)
 - d) option (iv)
4. Why the food is cooked quickly in pressure cooker? [1]
 - a) Steam cooks food quickly
 - b) Water boils at low temperature
 - c) Steam doesn't go waste
 - d) Increased pressure, increases the boiling point
5. The force of friction between two bodies is [1]
 - a) perpendicular
 - b) parallel
 - c) inclined at 30°
 - d) inclined at 60°
6. A rectangular block on the top of a table is kept in such a manner that its various sides touch the table. In different situations, this block imposes- [1]
 - a) Different thrust and different pressure
 - b) Uniform thrust and uniform pressure
 - c) Different thrust and uniform pressure
 - d) Uniform thrust and different pressure
7. Sliding is replaced in most machines by [1]
 - a) Use of ball bearings
 - b) Using graphite
 - c) Using powder
 - d) Using lubricants
8. When action and reactions forces are equal and opposite, They don't cancel [1]

- a) Because they have unequal mass
b) Because they are in state of rest
c) Because they have same direction
d) Because they are acting on two different bodies

9. What will be the pressure when 30 N force is applied on 120 cm² area? **[1]**
a) 2.5×10^2 N/m²
b) 5×10^4 N/m²
c) 2.5×10^3 N/m²
d) 5×10^3 N/m²

10. A box weighing 2 kg exerts a force of 20 N on the ground. The box covers an area of 2 sq. m on the ground. The pressure exerted by the box on the ground is **[1]**
a) 40 Pa
b) 30 Pa
c) 10 Pa
d) 20 Pa

11. How much force is applied on a body of mass 10 kg to increase its acceleration by 2m/s sq? **[1]**
a) 20 N
b) 10 N
c) 5 N
d) 12 N

12. While walking on the ground we apply force on **[1]**
a) friction
b) ground
c) gravitational
d) shoes

13. Pressure exerted by water at the bottom of the container depends on the **[1]**
a) Size of the column
b) All of these
c) Mass of the column
d) Height of the column

14. Ball bearings are used in bicycles, cars, etc. because **[1]**
a) the effective area of contact between the wheel and axle is reduced
b) the effective area of contact between the wheel and axle is increased
c) None of these
d) the actual area of contact between the wheel and axle is increased

15. Two forces each of 20 N act simultaneously on a body, the maximum net force and minimum net force are **[1]**
a) 40 N and 0
b) 20 N and 10 N
c) 20 N and 0
d) 40 N and 20 N

16. Friction between any two objects is due to **[1]**
a) Repulsion between them
b) Surface irregularities on them
c) Even surfaces of them
d) Attraction between them

17. A body will remain in state of motion on rolling on ground if **[1]**
a) Friction is very large
b) Friction is absent
c) All of the above
d) Friction is acting on same direction

18. The pressure on earth will be less when the man is **[1]**
a) standing by one foot
b) lying

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| | c) standing by two feet | d) sitting | |
| 19. | The force required to move a stationary body is called | | [1] |
| | a) Gravitational force | b) Unbalanced force | |
| | c) Frictional force | d) Balanced force | |
| 20. | The force by which earth attracts the other body is called | | [1] |
| | a) Contact force | b) Muscular force | |
| | c) Gravitational force | d) Magnetic force | |
| 21. | Pressure is | | [1] |
| | a. scalar quantity | | |
| | b. vector quantity | | |
| | c. Both (a) and (b) | | |
| | d. None of these | | |
| | a) Option (c) | b) Option (b) | |
| | c) Option (a) | d) Option (d) | |
| 22. | If no force acts on a body, it will | | [1] |
| | a) shape gets deformed | b) stop moving | |
| | c) either in rest or in motion | d) move fastly | |
| 23. | In game of tug-of war, two teams | | [1] |
| | a) Pull the rope tied with tree | b) Pull the rope in opposite direction | |
| | c) Pull the rope in same direction | d) Push the rope in opposite direction | |
| 24. | 1 horsepower is equal to- | | [1] |
| | a) 846 W | b) 746 W | |
| | c) 776 W | d) 11000 V | |
| 25. | Kabaddi players rub their hands with soil for a | | [1] |
| | a) Better grip of their opponents | b) Make the opponents weaker | |
| | c) Make the body more slippery | d) Increasing energy of the body | |
| 26. | A force creates acceleration of 10 m/s^2 in a body of 0.5 kg. The value of force will be: | | [1] |
| | a) 5 N | b) 6 N | |
| | c) 20 N | d) 50 N | |
| 27. | Truck or motor-bus has much wider tyres to | | [1] |
| | a) Increase pressure on road | b) Decrease the weight of truck | |
| | c) Decrease pressure on road | d) Decrease the energy consumption | |
| 28. | The action-reaction forces | | [1] |
| | a) act long different lines | b) act on different bodies | |
| | c) act on the same body | d) act in the same direction | |

29. A perpendicular force of 50 N is acting on a circular plate of radius 7cm. The pressure acting on plate will be [1]
 a) 10 Pa b) 100 Pa
 c) 1000 Pa d) 3246.75 Pa
30. A metal cuboid of mass M kg rests on a table surface are of 40 cm² in contact with the table. The pressure exerted by the cuboid on the table surface is 10000 Pa. Value of M is (given that $g = 10 \text{ ms}^{-2}$) [1]
 a) 5 kg b) 4 kg
 c) 2.5 kg d) 20 kg
31. The food can be cooked easily in pressure cooker, because- [1]
 a) Boiling happens at higher temprature b) Pressure remains constant
 c) Pressure change constantly d) Boiling happens at lower temperature
32. One Pascal is equal to [1]
 a) Gm per sq. cm b) N per sq. m
 c) Rm per sq. cm d) Hg per sq. m
33. **A ball is moving on a floor (surface) along a straight path with a uniform speed** State the conclusion you can draw from this statement [1]
 a) No gravitational force is acting on the bail. b) The force acting on the bail is zero.
 c) No frictional force acts on the rolling objects. d) A non-contact force is acting on the hall against frictional force.
34. During dry weather, while combing hair, sometimes we experience hair flying apart. The force responsible for this is [1]
 a) Force of friction b) Electrostatic force
 c) Force of gravity d) Magnetic force
35. The force depends on [1]
 a) Pressure and mass b) Density and acceleration
 c) Mass and acceleration d) Weight and acceleration
36. Relation between the SI unit of force and the weight of a 1 kg mass is [1]
 a) 1 kgf = 1 N b) 1 kgf = 9.8 N
 c) 1 N = 9.8 kgf d) 1 kgf = 0.98 N
37. S.I. unit of pressure is [1]
 a) Kelvin b) Celsius
 c) Newton d) Pascal
38. The soles of shoes are made up of groves to [1]
 a) it is very light b) the earth attracts it
 c) the air exerts a frictional force on it d) increase friction force acts on it
39. A coin flicked across a table stops because [1]

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| | a) it is very heavy | b) no force acts on it | |
| | c) the table exerts a frictional force on it | d) the earth attracts it | |
| 40. | At high altitudes atmospheric pressure | | [1] |
| | a) Is always zero | b) Increases | |
| | c) Remain same | d) Decreases | |
| 41. | Rate of change in velocity is called | | [1] |
| | a) Momentum | b) Conservation | |
| | c) Acceleration | d) Gravitation | |
| 42. | The weight of air in a column of the height of the atmosphere and area $10\text{cm} \times 10\text{ cm}$ is equal to | | [1] |
| | a) 100 kg | b) 1000 kg | |
| | c) 58 kg | d) 9.8 kg | |
| 43. | The lateral pressure exerted by liquid at same height is | | [1] |
| | a) Less in right direction | b) Equal in all direction | |
| | c) More in left direction | d) Not equal in all direction | |
| 44. | Liquid pressure increases with | | [1] |
| | a) Increase with atmospheric pressure | b) Decrease with air pressure | |
| | c) Increase in depth | d) Decrease in depth | |
| 45. | A force of a given magnitude acts on a body. The acceleration of the body depends on the | | [1] |
| | a) density | b) shape | |
| | c) mass | d) volume | |
| 46. | A man is standing on a weighing machine in a lift. The reading of machine will be minimum when machine will move | | [1] |
| | a) When lift falls freely downward | b) With constant speed upward | |
| | c) With constant acceleration downward | d) With constant acceleration upward | |
| 47. | One atmospheric pressure is equal to | | [1] |
| | a) $2.5 \times 10^5 \text{ N/m}^2$ | b) $1.1 \times 10^8 \text{ N/m}^2$ | |
| | c) $1.01 \times 10^5 \text{ N/m}^2$ | d) $1.5 \times 10^6 \text{ N/m}^2$ | |
| 48. | Which one of the following forces is a contact force? | | [1] |
| | a) Magnetic force | b) Force of gravity | |
| | c) Electrostatic force | d) Force of friction | |
| 49. | Sometimes, car skids on rainy season | | [1] |
| | a) water reduces the friction between the road and the tyres | b) water increases the friction between the road and the tyres | |
| | c) it is not possible to apply brakes on a wet road | d) the friction between the brake shoes and the wheels is reduced | |

50. The pressure inside our body is [1]
 a) Less than atmospheric pressure b) More than atmospheric pressure
 c) Equal to the atmospheric pressure d) Absent
51. Muscular force: contact force :: magnetic force : ? [1]
 a) Surface force b) Non-contact force
 c) Contact force d) Molecular force
52. The threaded tyres of cars, trucks, and bus [1]
 a) Increase the velocity of vehicles b) Provide better grips with ground
 c) Increase the diameter d) Decrease friction of ground
53. 1N/ sq m is equal to [1]
 a) 1 Pa b) 1.5 Pa
 c) 0.1 Pa d) 10 Pa
54. Which one is not the unit of pressure? [1]
 a) dyne cm⁻² b) N-m⁻²
 c) pascal d) N-m⁻¹
55. On which principle rocket works? [1]
 a) None of these b) On mass conservation
 c) On heat conservation d) On momentum conservation
56. Two objects repel each other. This repulsion could be due to [1]
 a) magnetic force only b) electrostatic force only
 c) either a magnetic or an electrostatic force d) frictional force only
57. Friction force between two rough surfaces can be _____ by using lubricant. [1]
 a) increased b) decreased
 c) kept same d) removed
58. In preparing a chapatti from dough _____ force is applied. [1]
 a) Friction b) Magnetic
 c) Muscular d) Gravitation
59. The speed of a falling body increases continuously. This is because [1]
 a) the earth attracts it b) it is very light
 c) the air exerts a frictional force on it d) no force acts on it
60. Force by the virtue of bodies in contact is called [1]
 a) strong force b) weak force
 c) contact force d) non-contact force
61. Pressure is [1]

- a) Inversely proportional to force b) Directly proportional to mass
c) Inversely proportion to area of contact d) Directly proportional to area of contact
62. Kicking a ball with our legs is an example of force of [1]
a) gravitation b) muscular
c) friction d) magnetic
63. The CGS unit of force is [1]
a) Pascal b) Dyne
c) Watt d) Joule
64. School bags have broader strips to [1]
a) Increase pressure b) Decrease mass
c) Decrease pressure d) Increase grips
65. The motion imparted to objects is due to the [1]
a) Action of gravity b) Action of weight
c) Action of pressure d) Action of force
66. Two boys exert force of 40 N and 60 N in opposite direction on an object. The resultant force is [1]
a) 20 N b) 100 N
c) 40 N d) 240 N
67. In which of the positions given below, any person creates maximum pressure on earth? [1]
a) While standing on earth b) While standing on one leg
c) While standing in calm position d) While laying down on earth
68. A motorcycle with worn out treads on tyres is more likely to skid due to [1]
a) Smaller size of tyres b) Less weight of tyres
c) More friction d) Less friction
69. A body moving on leveled grounds need to apply constant force to [1]
a) Contact force b) Gravitational force
c) Overcome frictional force d) Magnetic force of earth
70. If the force applied on a moving body in direction of motion than [1]
a) Mass of the body increases b) Speed of the body increases
c) Speed of body decreases d) Mass of the body decreases
71. The pressure in the bottle filled with water is maximum [1]
a) at the top of the bottle b) in the middle of bottle
c) in the bottom of the bottle d) first at the top then in the middle
72. It is easy to walk on sand with flat shoes because [1]
a) Flat shoes are sand resistant b) Flat shoes are made of plastic
c) Flat shoes are lighter d) Flat shoes have larger surface area

73. A force of 5 N is applied on a body to create an acceleration of 2m/s^2 . The mass of the body is equal to [1]
- a) 1.5 kg b) 2 kg
c) 2.5kg d) 5 kg
74. The frictional force exerted by fluids is also called [1]
- a) Slag b) Smug
c) Rung d) Drag
75. Which of the following effect is not produced by force? [1]
- a) Change in direction b) Change in shape
c) Change in speed d) Change in mass

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